

# Evaluating Integrals

## Today's Goals

- ◆ You should be able to apply integral formulas for basic functions, and understand where they come from.
- ◆ You should be able to find antiderivatives of sums, differences, and constant multiples of functions.
- ◆ You should be able to verify that you've correctly found an antiderivative of a function by taking a derivative.

With the Fundamental Theorem of Calculus and our knowledge of derivatives, we can figure out some useful formulas for integration:

## Basic Indefinite Integrals

- ◆ When  $n \neq -1$  is a constant,  $\int x^n dx = \frac{x^{n+1}}{n+1} + C$
- ◆  $\int \frac{1}{x} dx = \ln|x| + C$
- ◆  $\int e^x dx = e^x + C$
- ◆ When  $b > 0$  is a constant,  $\int b^x dx = \frac{1}{\ln b} \cdot b^x + C$
- ◆  $\int \sin x dx = -\cos x + C$
- ◆  $\int \cos x dx = \sin x + C$
- ◆  $\int \sec^2 x dx = \tan x + C$
- ◆  $\int \sec x \tan x dx = \sec x + C$
- ◆  $\int \frac{1}{1+x^2} dx = \arctan x + C$
- ◆  $\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C \text{ or } -\arccos x + C$

1. Evaluate the following definite and indefinite integrals.

(a)  $\int \frac{5}{7x} dx$

### Solution

First, this is an indefinite integral, so it's really asking for all antiderivatives of  $\frac{5}{7x}$ . Let's rewrite this:  $\int \frac{5}{7x} dx = \int \frac{5}{7} \cdot \frac{1}{x} dx = \boxed{\frac{5}{7} \ln|x| + C}$ .

(b)  $\int_{-e}^{-1/e} \frac{5}{7x} dx$

### Solution

Using our answer to (a),

$$\begin{aligned}\int_{-e}^{-1/e} \frac{5}{7x} dx &= \frac{5}{7} \ln |x| \Big|_{-e}^{-1/e} \\ &= \frac{5}{7} \ln \left( \frac{1}{e} \right) - \frac{5}{7} \ln e \\ &= \boxed{-\frac{10}{7}}\end{aligned}$$

(c)  $\int (e^\pi + ey + \pi^y) dy$

### Solution

$$\int (e^\pi + ey + \pi^y) dy = \boxed{e^\pi y + \frac{e}{2} y^2 + \frac{\pi^y}{\ln \pi} + C}. \text{ (Keep in mind that } e^\pi \text{ is a constant.)}$$

(d)  $\int (\pi + x)\sqrt{x} dx$

### Solution

We don't have a rule about integrating products, so let's first rewrite the integrand so that it's no longer a product:

$$\begin{aligned}\int (\pi + x)\sqrt{x} dx &= \int (\pi\sqrt{x} + x\sqrt{x}) dx \\ &= \int (\pi x^{1/2} + x^{3/2}) dx \\ &= \boxed{\frac{2}{3}\pi x^{3/2} + \frac{2}{5}x^{5/2} + C}\end{aligned}$$

(e)  $\int \left( \frac{e}{x^2} + 2^x \cdot 3^x \right) dx$

### Solution

We don't have a rule about integrating products, so we must first rewrite  $2^x \cdot 3^x$ :

$$\begin{aligned}\int \left( \frac{e}{x^2} + 2^x \cdot 3^x \right) dx &= \int (ex^{-2} + 6^x) dx \\ &= -ex^{-1} + \frac{1}{\ln 6} 6^x + C \\ &= \boxed{-\frac{e}{x} + \frac{1}{\ln 6} 6^x + C}\end{aligned}$$

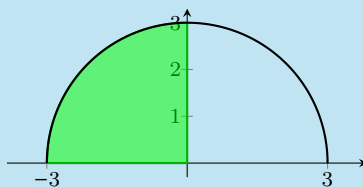
(f)  $\int_{-3}^0 (\sqrt{9-x^2} + x^2) dx$

### Solution

$\int \sqrt{9-x^2} dx$  is not one of the basic indefinite integrals that we know; however, you should recognize  $\sqrt{9-x^2}$  as a function whose graph is simple geometrically: it's a semicircle. So, let's split the integral up and use two different strategies:

$$\int_{-3}^0 (\sqrt{9-x^2} + x^2) dx = \int_{-3}^0 \sqrt{9-x^2} dx + \int_{-3}^0 x^2 dx.$$

We can visualize  $\int_{-3}^0 \sqrt{9-x^2} dx$  as the area of a quarter circle:



This area is  $\frac{9\pi}{4}$ . On the other hand,  $\int_{-3}^0 x^2 dx = \frac{1}{3}x^3 \Big|_{-3}^0 = 9$ . So,  $\int_{-3}^0 (\sqrt{9-x^2} + x^2) dx =$

$$\boxed{\frac{9\pi}{4} + 9}.$$

2. A busy bagel shop starts making bagels at 5 am each morning. One day, the rate at which bagels are made is

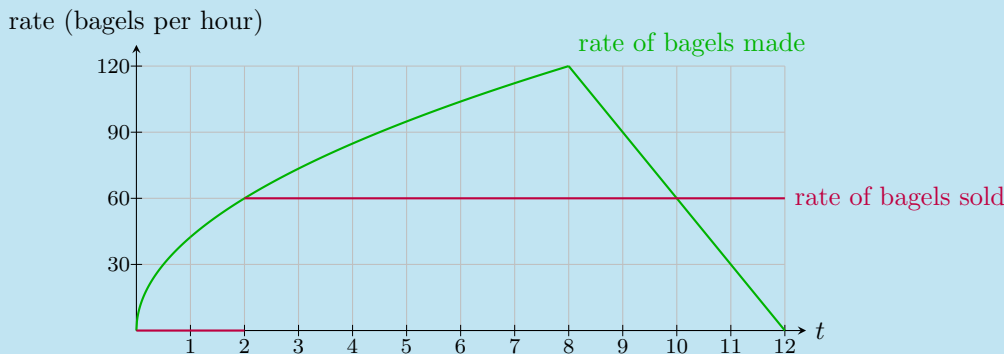
$$r(t) = \begin{cases} 30\sqrt{2t} & 0 \leq t \leq 8 \\ 30(12-t) & 8 < t \leq 12 \end{cases} \text{ bagels per hour,}$$

where  $t$  is measured in hours after 5am. The bagel shop is open from 7 am–5 pm; on this day, customers come at a steady rate of 60 customers per hour between 7 am and 5 pm. Each customer orders one bagel and receives it immediately.

- (a) At what times on this day is the number of bagels the shop has available increasing? At what times is it decreasing?

### Solution

First, here's the graph of  $r(t)$ , together with the rate at which bagels are sold:



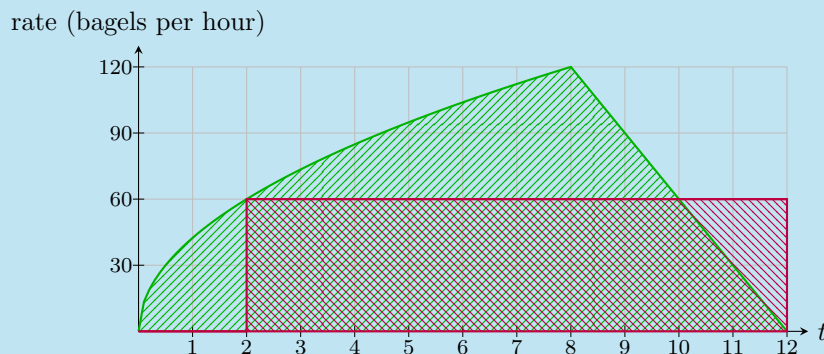
The number of bagels increases as long as bagels are made more quickly than they are sold;

therefore, the number of bagels is increasing before  $t = 10$  (3 pm) and decreasing after that.

- (b) At the end of the day, any unsold bagels are donated to a local food pantry. How many bagels are donated at the end of this day?

### Solution

The number of unsold bagels is (number of bagels made)  $-$  (number of bagels sold). We can picture the number of bagels made as the area under  $r(t)$  from  $t = 0$  to  $t = 12$  (the green area below); we can picture the number of bagels sold as the area under  $y = 60$  from  $t = 2$  to  $t = 12$  (the purple area below):



So, we want to calculate the green area minus the purple area. Notice that the green area from  $t = 8$  to  $t = 12$  is the same as the purple area from  $t = 8$  to  $t = 12$ , so we can just do (green area from  $t = 0$  to  $t = 8$ )  $-$  (purple area from  $t = 2$  to  $t = 8$ ). The green area from  $t = 0$  to  $t = 8$  is

$$\begin{aligned} \int_0^8 30\sqrt{2t} \, dt &= \int_0^8 30\sqrt{2}t^{1/2} \, dt \\ &= 20\sqrt{2}t^{3/2} \Big|_0^8 \\ &= 20\sqrt{2} \left( 8^{3/2} - 0 \right) \\ &= 20\sqrt{2} \cdot 8\sqrt{8} \\ &= 160\sqrt{16} \\ &= 640 \end{aligned}$$

The purple region from  $t = 2$  to  $t = 8$  is a rectangle of width 6 and height 60, so its area is 360. (That is, 360 bagels were sold.)

So, the number of unsold bagels at the end of the day is  $640 - 360 = \boxed{280}$ .

- (c) What is the largest number of bagels the shop has available at any point during this day?

### Solution

The number of bagels the shop has available is highest at  $t = 10$  (after all, we figured out in (a) that the number of bagels increases until  $t = 10$  and decreases after that). So, we need to

find the number of bagels at  $t = 10$ . This is

$$\begin{aligned} & (\text{number of bagels made by } t = 10) - (\text{number of bagels sold by } t = 10) \\ &= \int_0^{10} r(t) dt - (60 \text{ bagels per hour})(8 \text{ hours}) \end{aligned}$$

We found in (b) that  $\int_0^8 r(t) dt = 640$ , and we can see from the graph that  $\int_8^{10} r(t) dt = 180$ , so  $\int_0^{10} r(t) dt = 640 + 180$ :

$$\begin{aligned} &= 640 + 180 - 480 \\ &= \boxed{340} \end{aligned}$$

- (d) Are there times during the day when the shop has exactly 310 bagels available? If so, at how many times?

#### Solution

The shop starts with no bagels; the number of bagels increases to its maximum value of 340 at  $t = 10$  and then decreases to 280 at the end of the day, so it must have been 310 twice, once before  $t = 10$  and once after.

3. True or false:

- (a) If  $F$  is an antiderivative of  $f$  and  $G$  is an antiderivative of  $g$ , then  $F + G$  is an antiderivative of  $f + g$ .

#### Solution

True: if  $F' = f$  and  $G' = g$ , then  $(F + G)' = f + g$ .

- (b) If  $F$  is an antiderivative of  $f$  and  $G$  is an antiderivative of  $g$ , then  $F \cdot G$  is an antiderivative of  $f \cdot g$ .

#### Solution

False, and almost any example you try will show that this statement is false. For example,  $F(x) = x$  is an antiderivative of  $f(x) = 1$  and  $G(x) = 2x$  is an antiderivative of  $g(x) = 2$ , but  $F(x)G(x) = 2x^2$  is not an antiderivative of  $f(x)g(x) = 2$ .

4. (a) Decide whether each of the following definite integrals is positive or negative.

i.  $\int_{2\pi/3}^{3\pi/4} \frac{\cos^2 t - 1}{\cos^2 t} dt$

#### Solution

The function  $\frac{\cos^2 t - 1}{\cos^2 t}$  is always  $\leq 0$  (the numerator is always  $\leq 0$  because  $\cos t$  is between  $-1$  and  $1$ , and the denominator is always  $\geq 0$  because it's something squared). Since  $\frac{2\pi}{3} < \frac{3\pi}{4}$ , this integral must be negative.

ii.  $\int_0^{1/2} \left( \sqrt{\frac{1}{4} - x^2} + \frac{1}{\sqrt{1-x^2}} \right) dx$

### Solution

The functions  $\sqrt{\frac{1}{4} - x^2}$  and  $\frac{1}{\sqrt{1-x^2}}$  are always  $\geq 0$  (because  $\sqrt{\text{anything}}$  always refers to the positive square root). Since  $0 < \frac{1}{2}$ , this integral is positive.

(b) Now calculate the integrals, simplifying your answers.

### Solution

To evaluate  $\int_{2\pi/3}^{3\pi/4} \frac{\cos^2 t - 1}{\cos^2 t} dt$ , we rewrite

$$\begin{aligned} \int_{2\pi/3}^{3\pi/4} \frac{\cos^2 t - 1}{\cos^2 t} dt &= \int_{2\pi/3}^{3\pi/4} (1 - \sec^2 t) dt \\ &= t - \tan t \Big|_{2\pi/3}^{3\pi/4} \\ &= \left( \frac{3\pi}{4} + 1 \right) - \left( \frac{2\pi}{3} + \sqrt{3} \right) \\ &= \boxed{\frac{\pi}{12} + 1 - \sqrt{3}} \end{aligned}$$

To evaluate  $\int_0^{1/2} \left( \sqrt{\frac{1}{4} - x^2} + \frac{1}{\sqrt{1-x^2}} \right) dx$ , you should first recognize that we know an antiderivative of  $\frac{1}{\sqrt{1-x^2}}$  but not of  $\sqrt{\frac{1}{4} - x^2}$ , so we had better split this into two integrals:

$$\int_0^{1/2} \left( \sqrt{\frac{1}{4} - x^2} + \frac{1}{\sqrt{1-x^2}} \right) dx = \int_0^{1/2} \sqrt{\frac{1}{4} - x^2} dx + \underbrace{\int_0^{1/2} \frac{1}{\sqrt{1-x^2}} dx}_{=\arcsin x \Big|_0^{1/2}}$$

The graph of  $\sqrt{\frac{1}{4} - x^2}$  is the top half of the circle of radius  $\frac{1}{2}$  centered at the origin, so  $\int_0^{1/2} \sqrt{\frac{1}{4} - x^2} dx$  is the area of a quarter circle of radius  $\frac{1}{2}$ :

$$\begin{aligned} &= \frac{1}{4} \cdot \pi \left( \frac{1}{2} \right)^2 + \frac{\pi}{6} \\ &= \boxed{\frac{11\pi}{48}} \end{aligned}$$